CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 6

1 Ethene, C_2H_4, is an important molecule used in industry. It is an *unsaturated hydrocarbon*.

a Explain the terms *unsaturated* and *hydrocarbon*.

SKILLS PROBLEM SOLVING 9

b Ethene burns easily and releases a large amount of heat during combustion. Write a balanced chemical equation for the complete combustion of ethene.

SKILLS REASONING 7

c Explain why in industry ethene is considered too valuable to be used as a fuel.

SKILLS CRITICAL THINKING 6

2 Propene, C_3H_6, can be polymerised to make poly(propene).

a Explain the term *polymerisation*.

SKILLS INTERPRETATION 7

b Draw a displayed formula for propene.

8

c Draw a diagram to represent the structure of a poly(propene) chain showing three repeat units.

SKILLS CRITICAL THINKING 6

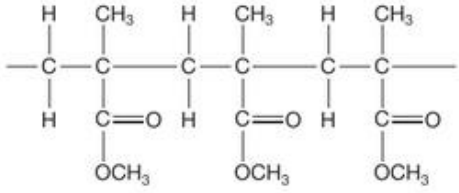
d The formation of poly(propene) is an example of *addition* polymerisation. Explain what is meant by the word *addition*.

SKILLS PROBLEM SOLVING 9

e Styrene has the formula $C_6H_5CH{=}CH_2$. Write a balanced chemical equation to show what happens when styrene is polymerised to make polystyrene. Your equation should clearly show the structure of the polystyrene. (Show the C_6H_5 group as a whole.)

SKILLS INTERPRETATION 8

f A small part of a Perspex molecule looks like this:



Draw the structure of the monomer from which Perspex is made.

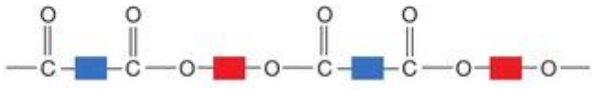
SKILLS DECISION MAKING 7

g Many addition polymers, for example those used for making plastic bags, are not biodegradable. Used plastic bags can be either buried underground in landfill sites or incinerated. State which method of disposal you think is better and give two reasons for your choice.

CHEMISTRY ONLY

SKILLS ANALYSIS

3 A polymer has the structure shown below:



a Identify the type of the polymer.

SKILLS INTERPRETATION

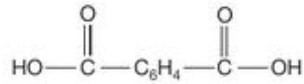
b Draw the structures of the monomers that are used to make this polymer.

SKILLS INTERPRETATION 8

4 Polyesters such as Terylene (for clothes) or PET (commonly used to make drinks bottles) are made by condensation polymerisation from ethane-1,2-diol and terephthalic acid (properly known as benzene-1,4-dicarboxylic acid). The structures of these are:

Ethane-1,2-diol: $HO–CH_2CH_2–OH$

Terephthalic acid:



a Draw a chain of the polymer with two repeat units.

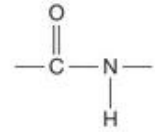
SKILLS PROBLEM SOLVING 9

b Write a balanced chemical equation for the formation of PET.

(The structure of the C_6H_4 group is complicated, and you can write it simply as C_6H_4. You can draw this as a block diagram if you wish, but it is more satisfying (and no more difficult in this case) to draw the structure properly.)

SKILLS CRITICAL THINKING 7

5 *(This question contains new material and is designed to look difficult. It is actually not too difficult as long as you understand about polyester.)* Nylon-6,6 is made by a condensation polymerisation of the monomers 1,6-diaminohexane ($H_2NCH_2CH_2CH_2CH_2CH_2CH_2NH_2$) and hexane-1,6-dioic acid. The amine group $–NH_2$ reacts in a very similar manner to the –OH group in an alcohol during condensation polymerisation with carboxylic acids, forming amide functional groups which join the monomers together into a polymer chain.



a i Explain what is meant by *condensation polymerisation* and how it differs from *addition polymerisation*.

SKILLS INTERPRETATION

ii Using a block diagram, draw one repeat unit for nylon-6,6.

SKILLS ANALYSIS, REASONING 8

b Nylon-6,10 is made from 1,6-diaminohexane and a longer chain acid, decanedioic acid, containing a total of 10 carbon atoms: $HOOCCH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2COOH$.

i How will a chain of nylon-6,10 differ from one of nylon-6,6? Refer to the diagram you drew in **a ii**.

ii In what way(s) will the two chains be the same? Again, refer to the diagram you drew in **a ii**.

END OF CHEMISTRY ONLY